

Guideline Title: Congestive Heart Failure & Pulmonary Edema

Guideline Number: GUID-3203-C005

Issue date: 09/09/2014	Replaces Dept. Policy:
Revision dates: 4/30/24	Developed by:
Approved by: Dr. Christopher Hunter, MD Air Care Team Medical Director	Approved by:
Signature: 	Signature:
Department Numbers: 3203	

- I. **PURPOSE:** Obtain and maintain sufficient cardiac output to maintain perfusion and oxygenation during acute heart failure.
- II. **DEFINITIONS:** None
- III. **DEPARTMENT GUIDELINE:**
 - a. Assess and manage airway, breathing, and circulation
 - i. Lung sounds for evidence of pulmonary edema (rales/wheezes).
 - ii. Auscultation of heart tones – S3 gallop may be present
 - iii. Obtain Ejection Fraction historical data if possible.
 - b. Provide supplemental oxygen and assist ventilations as needed to reduce work of breathing. Consider Non-Invasive Positive Pressure Ventilation - trial of CPAP or BiPAP if at a sending facility per GUID3203-PR018 *Non-Invasive Positive Pressure Ventilation*.
 - c. If necessary, secure an advanced airway per GUID3203-G008 *Rapid Sequence Induction for Endotracheal Intubation* and GUID3203-PR020 *Endotracheal Intubation* and ventilate with a transport ventilator per GUID3203-PR024 *Ventilation with a Mechanical Ventilator*.
 - d. Administer **Nitroglycerine (NTG) 0.4 mg SL every 5 minutes x 3 doses** if hemodynamically stable. Use caution in inferior wall MI or right ventricular infarct in V4R.
 - i. Consider initiating a **Nitroglycerin infusion at 10 mcg/min**, titrate until breathing improves, keeping MAP > 65 mmHg or SBP > 100 mmHg. Do not exceed 200 mcg/min.
 - ii. NTG is contraindicated if patient has received drugs for erectile dysfunction such as sildenafil (Viagra) or vardenafil (Levitra) within the past 24 hours and 72 hours for tadalafil (Cialis). (Consult medical direction).
 - e. Consider **Furosemide 40 mg IV**.
 - f. For pain management or comfort consider
 - i. **Morphine 0.1 mg/kg slow IV/IO** may repeat every 10 minutes PRN OR
 - ii. **Fentanyl 0.5-2 mcg/kg slow IV/IO** may repeat every 10 minutes PRN
 - iii. Monitor respiratory status and consider ETCO2 monitoring
 - g. If hypotension persists, consider initiating vasopressors per GUID3203-M011 *Hypotension*.

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- h. Consider inserting an indwelling urinary catheter to monitor urine output (If available at the sending facility and procedure does not unnecessarily delay transport).

IV. **DOCUMENTATION:** EMS Charts

V. **REFERENCES:**

- a. American Heart Association, 2015.